MLDM: Multiple-Location Disease Mapping

Statistical Model

$$\mathbf{Z}_{ij}|\mathbf{p}_{ij} \overset{\text{ind}}{\sim} \text{Multinomial}\left(m_{ij}, \mathbf{p}_{ij}\right),\ i=1,...,c;\ j=1,...,n_i;$$

$$p_{ijk} = \frac{\exp\left\{\beta_{ik} + \mathbf{x}_{ij}^{\top}\boldsymbol{\delta}_{ik} + \alpha_{ijk}\right\}}{1+\sum_{k'=2}^{d}\exp\left\{\beta_{ik'} + \mathbf{x}_{ij}^{\top}\boldsymbol{\delta}_{ik'} + \alpha_{ijk'}\right\}},\ k=1,...,d;$$

- $\mathbf{Z}_{ij} = \left(z_{ij1},...,z_{ijd}\right)^{\top}$, where z_{ijk} is the number of individual j's m_{ij} reported locations that fell within region k;
- $\mathbf{p}_{ij} = \left(p_{ij1},...,p_{ijd}\right)^{\top}$, $\sum_{k=1}^{d} p_{ijk} = 1$;
- Region $k=1$ is the reference category: $\beta_{i1} = \alpha_{ij1} = 0$ for all i,j, and $\boldsymbol{\delta}_{i1} = \mathbf{0}_{p_x}$ for all i;
- c: Number of disease groups; d: Number of regions; n_i: Number of individuals in group i; m_{ij}: Number of reported locations for individual j in group i;
- p_x: Length of individual-level covariate vector $\mathbf{x}_{ij}$ (no intercept; same length for all i,j);
- β_{ik}: Global, spatially-structured effect for disease group i in region k;
- $\boldsymbol{\delta}_{ik}$: Individual-level covariate effects for disease group i in region k;
- $\alpha_{ijk}|\sigma_i^2 \overset{\text{iid}}{\sim} \text{N}\left(0, \sigma_i^2\right)$: Individual-specific random effects.

Collect the non-reference parameters as $\boldsymbol{\theta}_k^{\top} = \left(\beta_{1k},...,\beta_{ck},\boldsymbol{\delta}_{1k}^{\top},...,\boldsymbol{\delta}_{ck}^{\top}\right)$ for $k=2,...,d$, and let $\boldsymbol{\theta}^{\top} = \left(\boldsymbol{\theta}_2^{\top},...,\boldsymbol{\theta}_d^{\top}\right)$. The joint prior distribution for $\boldsymbol{\theta}$ is:

$$\boldsymbol{\theta}|\boldsymbol{\gamma},\rho,\Omega \sim \text{MVN}\left(\mathbf{V}^*\boldsymbol{\gamma},\ \mathbf{Q}\left(\mathbf{W},\rho\right)^{-1}\otimes\Omega\right);$$

$$\mathbf{Q}\left(\mathbf{W},\rho\right) = \rho\left\{\text{diag}\left(\text{w}_{2\cdot},...,\text{w}_{d\cdot}\right) - \mathbf{W}\right\} + \left(1-\rho\right)\mathbf{I}_{d-1};$$

- $\mathbf{V}^*$: Stacked region-covariate design matrix (see manuscript for full definition);
- $\mathbf{W}$: $(d-1)\times(d-1)$ adjacency matrix; $\text{w}_{kk'} = 1$ if regions k and k' share a common border, $\text{w}_{kk} = 0$;
- $\text{w}_{k\cdot} = \sum_{k'=2}^{d}\text{w}_{kk'}$: Number of neighbors of region k;
- $\rho \in [0,1)$: Spatial correlation parameter; $\rho = 0$ implies spatial independence;
- Ω: $c(1+p_x)\times c(1+p_x)$ cross-covariance matrix;
- $\boldsymbol{\gamma}^{\top} = \left(\boldsymbol{\gamma}_1^{\top},...,\boldsymbol{\gamma}_{c(1+p_x)}^{\top}\right)$: Region-level covariate coefficients; $\boldsymbol{\gamma}_l^{\top} = \left(\gamma_{l1},...,\gamma_{lp_v}\right)$; p_v: Length of region-level covariate vector $\mathbf{v}_k$.

Prior Information

$\gamma_{lq} \overset{\text{iid}}{\sim} \text{N}\left(0,\sigma_{\gamma}^2\right),\ l=1,...,c(1+p_x);\ q=1,...,p_v.$

- Default setting: $\sigma_{\gamma}^2 = 10{,}000$.

$\Omega^{-1} \sim \text{Wishart}\left(\nu_{\Omega},\ \mathbf{S}_{\Omega}^{-1}\right).$

- Default setting: $\nu_{\Omega} = c(1+p_x)+1$, $\mathbf{S}_{\Omega} = \mathbf{I}_{c(1+p_x)}$.

$\rho \sim \text{Uniform}\left(0,1\right).$

$\sigma_i^2 \overset{\text{iid}}{\sim} \text{Inverse-Gamma}\left(a_{\sigma^2}, b_{\sigma^2}\right),\ i=1,...,c.$

- Default setting: $a_{\sigma^2} = 0.01$, $b_{\sigma^2} = 0.01$.

Default Initial Values

- $\boldsymbol{\theta}_k = \mathbf{0}_{c(1+p_x)}$ for all k;
- $\boldsymbol{\gamma} = \mathbf{0}_{p_v \cdot c(1+p_x)}$;
- $\Omega^{-1} = \mathbf{I}_{c(1+p_x)}$;
- $\rho = 0.50$;
- $\alpha_{ijk} = 0$ for all i, j, k;
- $\sigma_i^2 = 1.00$ for all i.

Notes

- MLDM_0 omits α_{ijk} and σ_i^2 from the model; use when $m_{ij} = 1$ for all i, j or as a simplified alternative to MLDM;
- `metrop_sd_rho_trans`: Standard deviation of the Normal proposal for ρ on the logit scale; must be specified by the user and tuned to achieve a reasonable acceptance rate.

Post-Hoc Functions

- `p_signal_space_create(beta)`: Computes $\widetilde{p}_{ik} = \exp\{\beta_{ik}\} / \left(1 + \sum_{k'=2}^{d} \exp\{\beta_{ik'}\}\right)$ from posterior samples of `beta`; returns a $c \times d \times T$ array;
- `p_signal_full_create(beta, gamma, v)`: Computes a covariate-adjusted version of $\widetilde{p}_{ik}$ that additionally removes the region-level covariate mean $\mathbf{v}_k^\top \boldsymbol{\gamma}_i$ from β_{ik} for $k = 2, ..., d$; returns a $c \times d \times T$ array;
- `psi1_create(p_signal)`: Computes $\psi_{1ik} = \widetilde{p}_{ik} - \frac{1}{c} \sum_{i'=1}^{c} \widetilde{p}_{i'k}$ from a `p_signal` array; returns a $c \times d \times T$ array;
- `psi2_create(p_signal)`: Computes $\psi_{2ik} = \widetilde{p}_{ik} - \frac{1}{d}$ from a `p_signal` array; returns a $c \times d \times T$ array.